ACIDS AND BASES 3

22

Concentration of acids and bases

Concentration is a measurement of how much of a solute is present in a solution. (Remember, **solute** is the substance being **dissolved**.) When dealing with solutions of acids and bases, chemists usually use moles per litre which is written as mol L^{-1} or M. Figure 1 shows how to calculate the number of moles of a substance. Figure 2 shows how to calculate the molarity of a solution.

$$\text{no. of moles} = \frac{\text{mass present (g)}}{\text{molar mass (g)}}$$

Figure 1

$$\text{concentration (moles per litre)} = \frac{\text{no. of moles}}{\text{volume (litres)}}$$

Figure 2

EXAMPLES

i **What is the molarity of a solution of hydrochloric acid when 0.35 moles are dissolved in 500 mL of water?**

Concentration = no. of moles ÷ volume

= 0.35 ÷ 0.5 = 0.70 mol L^{-1}

ii **What is the molarity of a solution made by dissolving 2.00 g of sodium hydroxide (NaOH) in 250 mL of water? (Atomic mass of Na = 23.00, O = 16.00, H= 1.00)**

Molar mass of sodium hydroxide = (23.00 + 16.00 + 1.00) = 40.00 g

Number of moles of sodium hydroxide = 2.00 g ÷ 40.00 g = 0.05 moles

Molarity = 0.05 ÷ 0.25 = 0.2 mol L^{-1}

1 Calculate the concentration of each solution in moles per litre.

a 2 moles of sulfuric acid in 1.5 litres of solution

b 1.75 moles of nitric acid in 600 mL of solution

c 12.5 g of hydrochloric acid (HCl) in 250 mL of solution (atomic mass of Cl = 35.5)

d 15.5 g of ammonia (NH_3) in 500 mL of solution (atomic mass of N = 14.00)

e 2.05 g of barium hydroxide ($Ba(OH)_2$) in 100 mL of solution (atomic mass of Ba = 137.3)

Concentration and strength

These two terms are often confused. **Concentration** refers to the amount of dissolved substance in a solution. A concentrated solution contains more **solute** than a dilute solution. **Strength** refers to the proportion of dissolved molecules that ionise when they dissolve in water. A strong acid has a higher percentage of molecules ionised in solution than a weak acid does.

Research 2 Give your own definitions of the terms *concentration* and *strength*, in respect to acids and bases. Include diagrams to illustrate your answers.

The pH scale Around 1909, S.P.L. Sørensen introduced the concept of pH while working for the Carlsberg brewery in Copenhagen. This is a way to describe hydrogen ion concentration (and hence acidity) in simple numbers. pH is the negative log (to the base ten) of the hydrogen ion concentration expressed in moles per litre. (See figure 3.) Note that logarithms use powers of ten to transform very large or very small numbers into more manageable units. For example, water is neutral. It has a pH of 7 which is a small, tidy number to work with. If the negative log wasn't used, we'd need to write the pH of water as 1.0×10^{-7} M. A pH of less than seven is **acidic**, a pH of seven is **neutral** and a pH above seven is **basic**.

$$\text{pH} = -\log_{10}[H^+]$$

Figure 3

3 Copy and complete the table.

$[H^+]$ (moles per litre)	pH	Type of solution
10^{-1}	1	
10^{-4}	4	
10^{-7}	7	
10^{-11}	11	
10^{-14}	14	

4 What relationship do you notice about the hydrogen ion concentration and the pH?

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Calculating pH

When carrying out calculations of pH on a scientific calculator, you will need to be familiar with the following operations, $\log_{10}$ (**log**, *not* ln), exponential (EXP or EE) and sign change (+/–).

Note that, depending on the calculator used, the sequence may differ from below.

EXAMPLES

iii What is the pH of a solution with hydrogen ion concentration 3.5×10^{-4} moles per litre?

Enter the following:

- LOG
- 3.5
- EXP
- 4
- +/–
- =
- +/– The answer is 3.45.

iv What is the pH of a solution with hydrogen ion concentration 0.05 moles per litre?

Enter the following:

- LOG
- 0.05
- =
- +/– The answer is 1.30.

5 Calculate the pH of each solution. Show your working.

a $[H^+] = 0.40$ moles per litre

b $[H^+] = 0.00001$ moles per litre

c $[H^+] = 3.0$ moles per litre

d $[H^+] = 3.2 \times 10^{-3}$ moles per litre

e $[H^+] = 9.2 \times 10^{-7}$ moles per litre

f $[H^+] = 3.2 \times 10^{-12}$ moles per litre

Why do we need to know pH values? pH can help chemists identify chemicals. Many chemical processes need to take place at a specific pH in order to obtain optimum results. Many agricultural crops need soil of a particular pH to give the best yield. Marine life is affected if pH changes due to a discharge of industrial waste. Human skin has a pH of 5.5 and many cosmetics and cleansers have been set to this value to minimise skin irritation. Cell function in the body is dependent on body pH levels being within a certain limit. Enzyme activity is vital for keeping body processes working properly and pH levels directly affect this activity.

6 Explain what is meant by *optimum results*.

7 What is *marine life*?

8 Define *pH*.

9 Explain what is meant by a pH of 5.5.

Measuring pH **Indicators** have long been a way of establishing the approximate pH of a solution. An indicator is a dye that changes colour over a specific pH range. Indicators can be in the form of solutions or strips of paper can be soaked in the indicator. One of the most well-known indicators is **litmus**. This is a dye (known for hundreds of years) that is obtained from lichens. Litmus turns red in acidic solutions and blue in basic solutions. **Universal indicator** is a mixture of several indicators which have an overlap of colour-change over a range of pH.

Research **10** Give the names of three different indicators (apart from litmus), their colours and the pH ranges at which they change colour.

Research **11** Draw up a table that shows the colours of universal indicator from pH 0 to pH 14.

pH can also be measured electronically. A pH meter consists of a special electrode connected to a meter. It is calibrated by immersing the electrode in solutions of known pH (buffers). pH meters can measure pH more accurately than chemical indicators and can also be connected to data loggers to plot changes in pH against variables such as time. They are more expensive and complex to use than chemical indicators.

12 Give two advantages and two disadvantages of pH meters compared to indicators.

Research **13** Buffer solutions have uses other than calibrating pH meters. Find out what buffer solutions are and what else they can be used for.

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